

DAY04:- TCP/IP & OSI MODEL

Name:- Rishabh Raj

Day:- 04

Internship Id:- JVL-CS-2026-A70F

1. What is a network protocol?

A network protocol is a set of rules that defines how devices communicate and exchange data over a network.

2. What is the OSI Model?

OSI (Open System Interconnection) Model is a 7-layer reference model used to understand how data is transmitted between network devices.

3. Name all seven OSI layer in correct order?

From Layer 7 to Layer 1:

1. Application
2. Presentation
3. Session
4. Transport
5. Network
6. Data Link
7. Physical

4. What is the purpose of the Application Layer?

It provides network services directly to applications used by users, such as HTTP, HTTPS, FTP, DNS and SMTP.

5. What is the purpose of the Transport Layer?

It provides end-to-end communication, reliability, flow control and segmentation of data. TCP and UDP operate at this layer.

6. What is the purpose of the Network Layer?

It handles logical addressing and routing of packets between different networks. IP addresses operate at this layer.

7. What is the purpose of the Data Link Layer?

It provides node-to-node communication, framing and MAC addressing. Ethernet Operates mainly at this layer.

8. What is the purpose of the Physical Layer?

It transmits raw bits through physical media such as cables, fiber and radio signals.

9. What is the TCP/IP Model?

The TCP/IP Model is the networking model used for communication over the Internet. It commonly has four layers:

1. Application
2. Transport
3. Internet
4. Network Access

10. Difference between OSI and TCP/IP models

OSI	TCP/IP
• 7 layers • Mainly a reference model • Developed by ISO • Session & Presentation are separate • More detailed	• 4 layers commonly • Practical Internet networking model • Developed around ARPA/ DoD networking • Included in Application layer • More practical and widely implemented

11. What is encapsulation?

Encapsulation is the process of adding headers to data as it moves down through the network layers before transmission.

12. What is decapsulation?

Decapsulation is the process of removing headers/trailers as received data moves up through the network layers toward the application.

13. Which layer uses IP addresses?

Network Layer (Layer 3).

14. Which layer uses MAC addresses?

Data Link Layer (Layer 2).

15. Which layer uses TCP and UDP?

Transport Layer (Layer 4).

16. Which layer includes HTTP and HTTPS?

Application Layer (Layer 7).

17. Why should cybersecurity professionals understand networking layers?

Cybersecurity professionals need to understand networking layers to analyze network traffic, identify attacks, troubleshoot security issues, monitor communication and understand where different security controls operate.

18. Explain how network data travels from a browser to a server.

When you enter a website in a browser:

1. Application Layer: Browser creates an HTTP/HTTPS request.
2. Transport Layer: TCP/UDP handles transport; HTTPS commonly uses TCP with TLS for traditional HTTPS.
3. Network Layer: IP adds source and destination IP addresses.
4. Data Link Layer: Ethernet/Wi-Fi uses MAC addresses and creates frames.
5. Physical Layer: Data is transmitted as electrical, optical or radio signals.
6. The server receives the data and decapsulates it layer by layer.
7. The server processes the request and sends a response back using the reverse process.

19. Your resolved IP address for google.com (nslookup)

```
[rishabhraj@Rishabhs-MacBook-Air ~ % nslookup google.com
Server:         2409:40d4:117:eddb::cf
Address:        2409:40d4:117:eddb::cf#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.29.102
Name:   google.com
Address: 142.250.29.101
Name:   google.com
Address: 142.250.29.139
Name:   google.com
Address: 142.250.29.100
Name:   google.com
Address: 142.250.29.113
Name:   google.com
Address: 142.250.29.138

rishabhraj@Rishabhs-MacBook-Air ~ %
```

20. Number of traceroute hops to google.com

```
C:\Users\satya>tracert google.com

Tracing route to google.com [2404:6800:4002:81d::200e]
over a maximum of 30 hops:

  1     5 ms     3 ms     2 ms  2409:40d4:117:eddb::cf
  2     *         *         *    Request timed out.
  3    77 ms    208 ms    337 ms 2405:200:5216:0:3925::1
  4   104 ms    111 ms     96 ms 2405:200:801:1600::780
  5     *         *         *    Request timed out.
  6    13 ms    18 ms    15 ms 2405:200:812:1513:61::7
  7     *         *         *    Request timed out.
  8     *         *         *    Request timed out.
  9     *         *         *    Request timed out.
 10     *         *         *    Request timed out.
 11    24 ms    18 ms     *    2001:4860:1:1::17ae
 12    31 ms    27 ms   163 ms 2001:4860:0:1::7819
 13    70 ms    40 ms    40 ms 2001:4860:0:1::5e49
 14    22 ms    36 ms    23 ms del11s18-in-x0e.1e100.net [2404:6800:4002:81d::200e]

Trace complete.
```

Screenshots

Ping google.com

```
rishabhraj@Rishabhs-MacBook-Air ~ % ping google.com
PING google.com (142.250.182.46): 56 data bytes
64 bytes from 142.250.182.46: icmp_seq=0 ttl=111 time=36.440 ms
Request timeout for icmp_seq 1
64 bytes from 142.250.182.46: icmp_seq=2 ttl=111 time=79.261 ms
64 bytes from 142.250.182.46: icmp_seq=3 ttl=111 time=38.023 ms
64 bytes from 142.250.182.46: icmp_seq=4 ttl=111 time=35.905 ms
64 bytes from 142.250.182.46: icmp_seq=5 ttl=111 time=47.456 ms
64 bytes from 142.250.182.46: icmp_seq=6 ttl=111 time=169.317 ms
Request timeout for icmp_seq 7
64 bytes from 142.250.182.46: icmp_seq=8 ttl=111 time=51.347 ms
64 bytes from 142.250.182.46: icmp_seq=9 ttl=111 time=56.072 ms
64 bytes from 142.250.182.46: icmp_seq=10 ttl=111 time=37.345 ms
64 bytes from 142.250.182.46: icmp_seq=11 ttl=111 time=30.833 ms
64 bytes from 142.250.182.46: icmp_seq=12 ttl=111 time=43.307 ms
64 bytes from 142.250.182.46: icmp_seq=13 ttl=111 time=39.023 ms
64 bytes from 142.250.182.46: icmp_seq=14 ttl=111 time=40.445 ms
Request timeout for icmp_seq 15
64 bytes from 142.250.182.46: icmp_seq=16 ttl=111 time=47.248 ms
64 bytes from 142.250.182.46: icmp_seq=17 ttl=111 time=47.798 ms
64 bytes from 142.250.182.46: icmp_seq=18 ttl=111 time=21.794 ms
64 bytes from 142.250.182.46: icmp_seq=19 ttl=111 time=72.754 ms
```

Tracert google.com

```
C:\Users\satya>tracert google.com

Tracing route to google.com [2404:6800:4002:81d::200e]
over a maximum of 30 hops:

  1      5 ms      3 ms      2 ms      2409:40d4:117:eddb::cf
  2      *          *          *          Request timed out.
  3     77 ms     208 ms     337 ms     2405:200:5216:0:3925::1
  4    104 ms     111 ms      96 ms     2405:200:801:1600::780
  5      *          *          *          Request timed out.
  6     13 ms     18 ms     15 ms     2405:200:812:1513:61::7
  7      *          *          *          Request timed out.
  8      *          *          *          Request timed out.
  9      *          *          *          Request timed out.
 10      *          *          *          Request timed out.
 11     24 ms     18 ms      *          2001:4860:1:1::17ae
 12     31 ms     27 ms     163 ms     2001:4860:0:1::7819
 13     70 ms     40 ms     40 ms     2001:4860:0:1::5e49
 14     22 ms     36 ms     23 ms     del11s18-in-x0e.1e100.net [2404:6800:4002:81d::200e]

Trace complete.
```

Nslookup google.com

```
rishabhraj@Rishabhs-MacBook-Air ~ % nslookup google.com
Server:      2409:40d4:117:eddb::cf
Address:     2409:40d4:117:eddb::cf#53
```

Non-authoritative answer:

```
Name:   google.com
Address: 142.250.29.102
Name:   google.com
Address: 142.250.29.101
Name:   google.com
Address: 142.250.29.139
Name:   google.com
Address: 142.250.29.100
Name:   google.com
Address: 142.250.29.113
Name:   google.com
Address: 142.250.29.138
```